

Dynamic Acousto-Caustics in Dual-Optimized Holographic Fields

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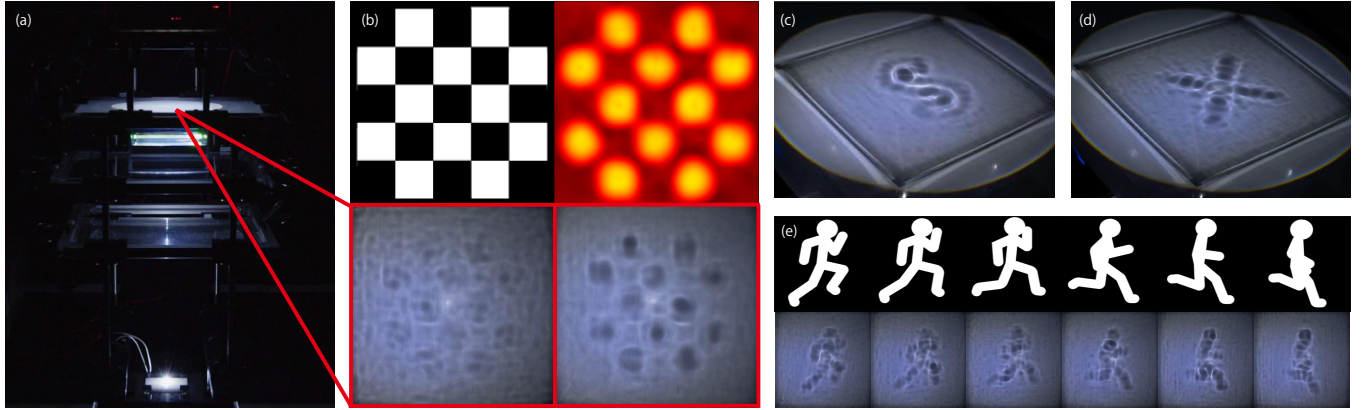


Figure 1: (a) The actual setup used. This setup, from top to bottom, consists of a PAT, mesh, a water tank with silicon oil, a Fresnel lens, and a light. The mesh serves as a screen to display the caustics. (b) Top left: target pattern; Top right: sound pressure distribution optimised for the target pattern; Bottom left: focus area before Digital Twin; Bottom right: focus area after Digital Twin. (c) ‘S’ written by focusing caustics. (d) Cross mark drawn in shadow. (e) A human running. Six frames are listed here. Top: These serve as the target images. Bottom: The actual animation of these target images as dynamic caustics.

ABSTRACT

We aim to computationally replicate the dynamic caustics observed in nature, such as the captivating light patterns found at the pool or within flowing rivers. Traditional computer science approaches have only been able to reproduce these light patterns through static fabrication. We introduce “Dynamic Acousto-Caustics” a method that merges acoustofluidics with optics to dynamically manipulate light caustics by controlling the shape of liquid surfaces. Using computer-controlled acoustic fields to deform the surface of a liquid medium, we generate dynamic light behaviors in fluid patterns unachievable with static refractive surfaces. This research not only extends the understanding and application of controlled caustics across various technical and creative domains but also exemplifies the intersections made possible at the convergence of multidisciplinary research. While devices exist to generate waves by vibrating

water surfaces, they could not produce continuous caustics at specific locations and timings. Employing ultrasonics to deform the water surface allows us to directly manipulate the surface to create more continuous animated patterns. We shape caustics through acoustic field manipulation and optimize the visual outcome emanating from the geometry of these manipulated objects within a dynamic system. Our approach leverages the Digital Twin methodology as an optimization strategy to fine-tune the interplay between sound and light. This enhances the understanding and application of controlled caustics in numerous technical, demonstrating the crossing points enabled by the convergence of interdisciplinary research streams.

ACM Reference Format:

Koki Nagakura, Tatsuki Fushimi, Ayaka Tsutsui, and Yoichi Ochiai. 2024. Dynamic Acousto-Caustics in Dual-Optimized Holographic Fields. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641517.3664384>

1 INTRODUCTION

Caustics refer to the distinctive patterns that occur when light reflects or refracts off an object’s surface, casting a projection onto

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SIGGRAPH Emerging Technologies '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0524-3/24/07

<https://doi.org/10.1145/3641517.3664384>

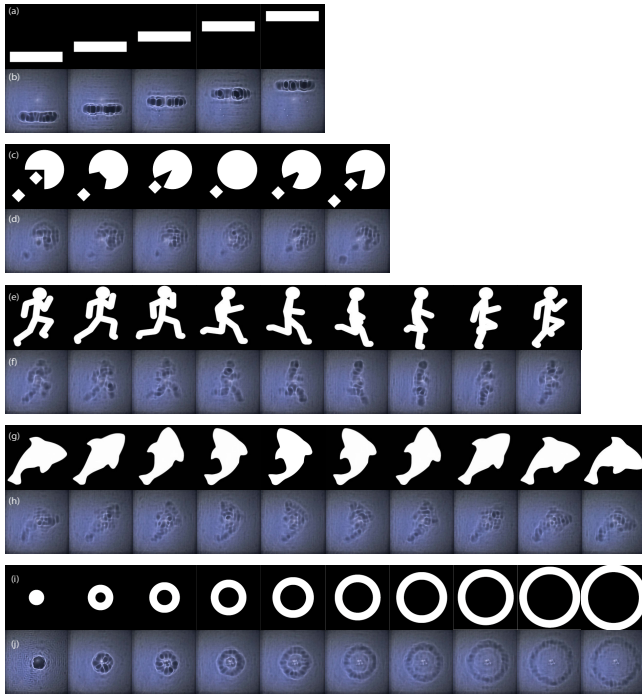


Figure 2: Comparison of the target image used for the animation and the generated caustics. Here, the caustics are presented frame by frame rather than in sequence as they were animated. The actual animation, which is generated by projecting these frames in succession, is included in the SUPPLEMENTAL MATERIAL. Each frame was created by superimposing the sound pressure of 9 frames to optimize the initial sound pressure distribution, followed by Digital Twin processing.

another surface. These phenomena exhibit dynamic and alluring light patterns that captivate observers, commonly witnessed in settings such as pools, flowing rivers, and seas. Our research aims to computationally reproduce these dynamic caustics found in natural environments.

We introduce an approach that merges acoustofluidics and optics to dynamically manipulate light caustics by controlling the shape of liquid surfaces, which we term “Dynamic Acousto-Caustics”. Traditional computer science approaches have facilitated the reproduction of caustic light patterns through static fabrication [Papas et al. 2011; Schwartzburg et al. 2014; Yue et al. 2014]. However, these studies frequently employed static solid surfaces as refractive media. The usage of liquid surfaces to form caustics has been proposed before [Suzuki et al. 2019]. Yet, this has been confined to in silico and has not been applied to real-world fluid manipulation.

Our research successfully utilizes ultrasonics to directly push and shape fluid surfaces, creating dynamic caustics capable of depicting text and imagery. Phased Array Transducers (PAT) have been enthusiastically adopted in computer graphics for applications such as touch display [Monnai et al. 2014] and aerial displays [Hoshi et al. 2010; Long et al. 2014]. Our approach does not solely

rely on the optimization of acoustic fields using holograms but involves shaping caustics by utilizing an acoustic field to generate the manipulated object and optimizing the resultant light pattern from this object’s geometry. We utilized the Digital Twin methodology [Fushimi et al. 2024] as an optimization strategy for fine-tuning the interplay between sound and light.

2 CAUSTICS GENERATION MECHANISM

The ultrasonic waves emitted from the PAT pass through the mesh structure screen and generate sound pressure on the surface of the silicone oil in the tank. The deformed surface refracts the parallel light passing through the fresnel lens, forming a pattern on the screen. In order to further enhance the quality of the generated caustics, we perform experimental optimization. Detailed experimental setups are available in the Supplemental Material.

3 EXPERIENCE

The installation allows visitors to directly observe ‘dynamic acoustics’, achieved through the fusion of acoustics and optics. By using a computer-controlled sound field to deform the surface of the liquid medium, dynamic light behaviour is generated in the fluid pattern that cannot be achieved with static refractive surfaces. Participants can view the resulting caustics either directly on the screen of the experimental setup or on a real-time video displayed on the screen and captured by a camera. The displayed dynamic caustics include various animation patterns prepared here (e.g. running stickmen, ripples, etc.) and a demonstration of caustics optimisation using a Digital Twin.

ACKNOWLEDGMENTS

We would like to extend our sincere gratitude to Sony Semiconductor Solutions Corporation for their generous support in lending us a sensor kit which was instrumental in the completion of our research.

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